# The genome of the Pacific acorn barnacle provides insights into the evolution of extremely large populations

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**Abstract**

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## Introduction

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## Results

### Genome assembly

We assembled a chromosome-level reference genome for the Pacific acorn barnacle using PacBio Hi-Fi long reads and Hi-C chromosome conformation capture. Our initial contig-level assembly, generated using `hifiasm` [Cheng2021; Cheng2022] with default parameters, generated sequences with a total length of 1.64 Gbp, composed of 4,212 contigs, and a contig N50 of 734 kbp. We also assessed gene-completeness using `BUSCO`s [Manni2021; Tegenfeldt2024], as implemented in `compleseam` [Huang&Li2023], resulting in a completeness of 95.66% (`S`:45.31%; `D`:50.35%; `F`:0.79%; `M`:3.55%; `N`:1,013) for the `arthropoda_odb10` single-copy ortholog set. The statistics for this initial assembly, with twice the expected genome size and over 50% duplicated `BUSCO`s, suggested the presence of a large fraction of haplotig sequences, reflecting the high heterozygosity expected for the species.

Following optimzation, both in the integration of additional data types and in the increased stringency of the purging of haplotig and contaminant sequences (see Methods), we generated a contig-level assembly composed of 1,342 sequences, with a total length of 1.05 Gbp, a contig N50 of 1.35 Mbp, and a `BUSCO` completeness of 93.88% ([`S`:89.04%; `D`:4.84%; `F`:1.18%; `M`:4.94%; `N`:1,013], `arthropoda_odb10`). The contig-level assembly was then scaffolded with Hi-C data using the `YAHS` software [Zhou2022]. Our scaffolding process generated an assembly with a total length of 1.05 Gbp, composed of 650 scaffolds, a scaffold N50 of 50.96 Mbp, and a `BUSCO` completeness of 93.98% ([`S`:89.24%; `D`:4.74%; `F`:0.89%; `M`:5.13%; `N`:1,013], `arthropoda_odb10`).

Table 1: Assembly statistics for the *Balanus glandula* genome at different stages of the assembly process.

| Assembly Stage | Total Length (Gbp) | N50 (Mbp) | BUSCO (%) |
|---|---|---|---|
| Default contigs | 1.64 | 0.734 | 95.66 |
| Optimized contigs | 1.05 | 1.35 | 93.88 |
| Hi-C scaffolds | 1.05 | 50.96 | 93.98 |
| Curated assembly | 1.04 | 51.41 | 94.08 |

This scaffold-level assembly was further processed by performing manual curation of the Hi-C contact map and self-correction using the software `Inspector` [Chen2021]. This curated assembly was composed of 592 fragments, a total sequence length of 1.04 Gbp, a scaffold N50 of 51.41 Mbp, and a per-base quility value (QV) of 30.49. When assessing completeness, the curated assembly was 94.08% (`S`:89.44%; `D`:4.64%; `F`:0.89%; `M`:5.03%; `N`:1,013) and 92.77% (`S`:87.77%; `D`:5.40%; `F`:2.93%; `M`:4.30%; `N`:1,536) gene-complete for the `arthropoda_odb10` and `crustacea_odb12` `BUSCO` datasets, respectively. Of the 1.04 Gbp of total sequence, 906.6 Mbp (86.74%) was contained in 58 scaffolds larger than 1 Mbp. Moreover, the 16 largest sequences (larger than 8.5 Mbp) contain 841.2 Mbp (80.61%) of the total sequence, and likely represent 16 putative chromosomes. This karyptype number, N=16, is in agreement with that observed for the genus *Balanus* [Austin1958] and other acorn barnacle species in the family Balanidae [Han2024].

## Genome annotation

Once the assembly and scaffolding was completed, we annotated and masked repeat sequences in the *B. glandula* reference assembly using the `EarlGrey` software [Baril2024]. This analysis identified 698.6 Mbp (66.94%) of the assembly as repetitive elements; however, the majority of these, 404.5 Mbp (38.76% of the assembly), remain unclassified among the major repeat element types. This large proportion of unclassified repeat elements is likely the product of the very small number of representative barnacle sequences among curated repeat reference libraries, reflecting barnacle's status as a genomically understudied group. Among the classified repeats, DNA and LINE elements were the two most abundant repeats classes, comprising 8.45% and 9.16% of the total assembly, respectively.

Following repeat annotation and masking of the reference genome, we annotated protein-coding genes using short-read RNAseq and long-read PacBio Iso-Seq sequencing, combining and curating annotations generated across the two data types. This process resulted in 20,234 protein-coding genes annotated in the genome. The resulting annotation was 93.58% (`S`:79.76%; `D`:13.82%; `F`:1.78%; `M`:4.64%; `N`:1,013) and 84.77% (`S`:72.66%; `D`:12.11%; `F`:4.10%; `M`:11.13%; `N`:1,536) gene-complete when compared against the `arthropoda_odb10` and `crustacea_odb12` `BUSCO` datasets, respectively. Additionally, we compared our annotation against the proteome of four other barnacle species (see Methods) using `OrthoFinder` [Emms&Kelly2015; Emms&Kelly2019], resulting in the assignment of 16,978 (83.91%) *B. glandula* genes to orthologous groups across barnacles and the identification of 2,576 single-copy orthologs across all five species.

In addition to protein-coding genes, we also annotated non-coding elements in the *B. glandula* reference assembly. We identified 7,532 long, non-coding RNAs (lncRNA), 1,203 transfer RNAs (tRNA), and 24 ribosomal RNA (rRNA) genes. In total, across protein-coding and non-coding transcripts, we were able to annotate 28,993 genes in the genome.

## Conserved architecture across barnacle genomes

We performed a conserved synteny analysis using the software `Genespace` [Lovell2022] to identify patterns of large-scale structure across barnacle genomes. We compared our *B. glandula* genome against the chromosome-level genome assemblies of *Amphibalanus amphitrite* (Balanidae) [Han2024] and *Pollicipes pollicipes* (Pollicipedidae) [Bernot2022]. These genomes display consevation in their large-scale organization, including correspondance between 16 one-to-one orthologous chromosomes across the three species. This finding both shows that our scaffolding of the *B. glandula* assembly likely reflects the biological structure of the genome and is not the product of technical artifacts, but also indicates the conservation in the genome architecture of barnacle genomes across 1̃65 Mya of evolution [Han 2024; Bernot2022].

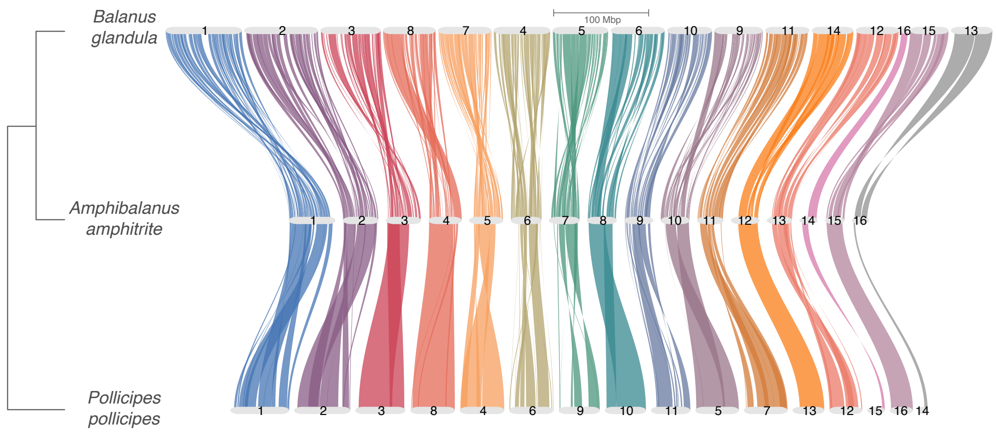


Figure 1: Conserved synteny across barnacle genomes. Genome-wide riparian synteny plots showing one-to-one correspondence across 16 orthologous chromosomes between the genomes of *Balanus glandula* (top), *Amphibalanus amphitrite* (middle), and *Pollicipes pollicipes* (bottom). Each set of colored lines shows orthologous genes across the three assemblies, colored according to their chromosome of origin. Chromosomes are drawn according their order in *A. amphitrite*. Cladogram on the left shows the evolutionary relationship among the three species.

## Discussion

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## Materials and Methods

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### Data Availability

https://github.com/arcolon14/balanus_genome_ms

## Acknowledgments

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# Supplements